Week 1

Introduction to Probability

Paul Laskowski and Alex Hughes
UC Berkeley, School of Information

Speaker notes

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Speaker notes

No notes on this slide.

The Nature of Statistical Models

Speaker notes

No notes on this slide.

What is a Model?

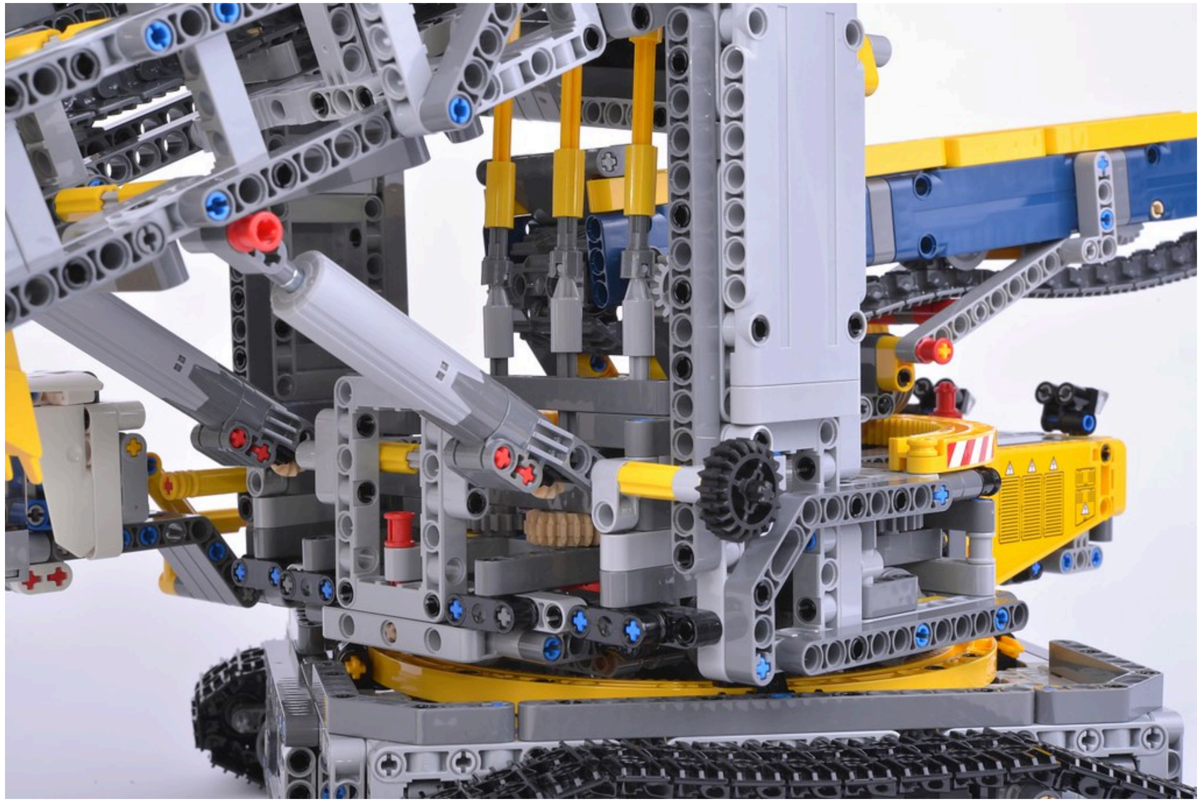


Image: www.flickr.com/photos/brickset/ (CC BY 2.0)

Speaker notes

- This course is about statistical modeling. So let's start by asking, what is a model?
- Here's a model, built out of lego bricks.
- Some of you might have played with legos growing up, I definitely did.
- The thing with legos is you learn some basic rules: the ways that legos can fit together.
- Then you can build something that resembles something in the real world, that captures important behavior.
- When you're starting out with legos for the first time, you don't immediately try to build a giant machine.
- Maybe you take two legos, test them to see how they fit together. Once you're familiar with the pieces, you can then try to create something that you see in the world.

What is a Statistical Model?

HTTHTHHTTH….

Speaker notes

- A statistical model is also a representation of something in the world.
- When you have data, it comes from a real-world system.
- But we don’t just care about this data, we have questions: what data is coming next? what other data could have plausible come up? what features of the world are consistent with this data?
- For those questions, we can’t interrogate the world directly, we need a statistical model.
- This is a representation that also creates data.
- We have to pretend that this data actually came from the model. Then we can start to answer these questions.

Course Plan

1. Putting together statistical models
2. Fitting models to data

Speaker notes

- When we designed this course, we decided to follow a very strict narrative.
- At the highest level of abstraction, here’s our plan…
- Most courses jump around, looking at data, then building models, then back.
- We believe it’s better to really commit to understanding models, then apply them.
- Because of that, we’re actually not going to touch any data for several weeks.
- But don’t worry, there will be a LOT of application in the second half of the course.

First Models

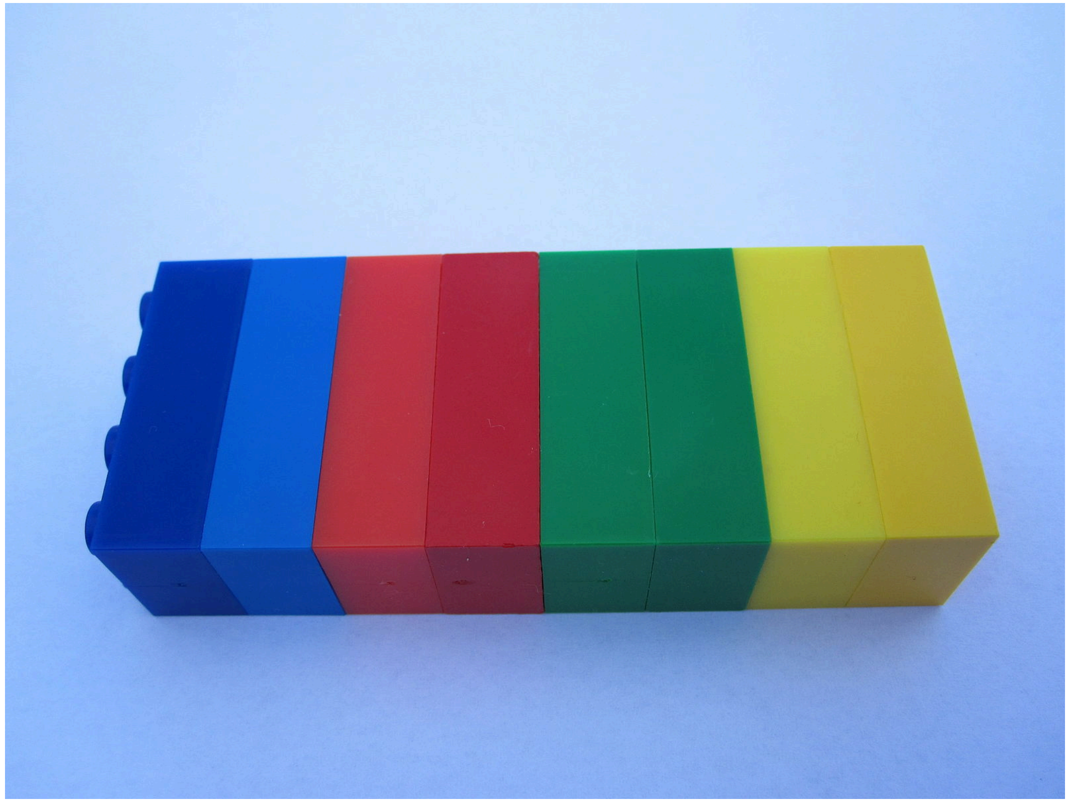


Image: Hans Schou (CC BY-SA 3.0)

Speaker notes

- When we start out, our models are not going to resemble the world at all
- They will be brittle - with no flexibility to make them look like the real world
- Actually, just like this tower of legos, they may be very rectangle.
- You may ask, where did this rectangular distribution come from?
- The answer is we made it up! and we just want to build something simple to understand how the bricks fit together

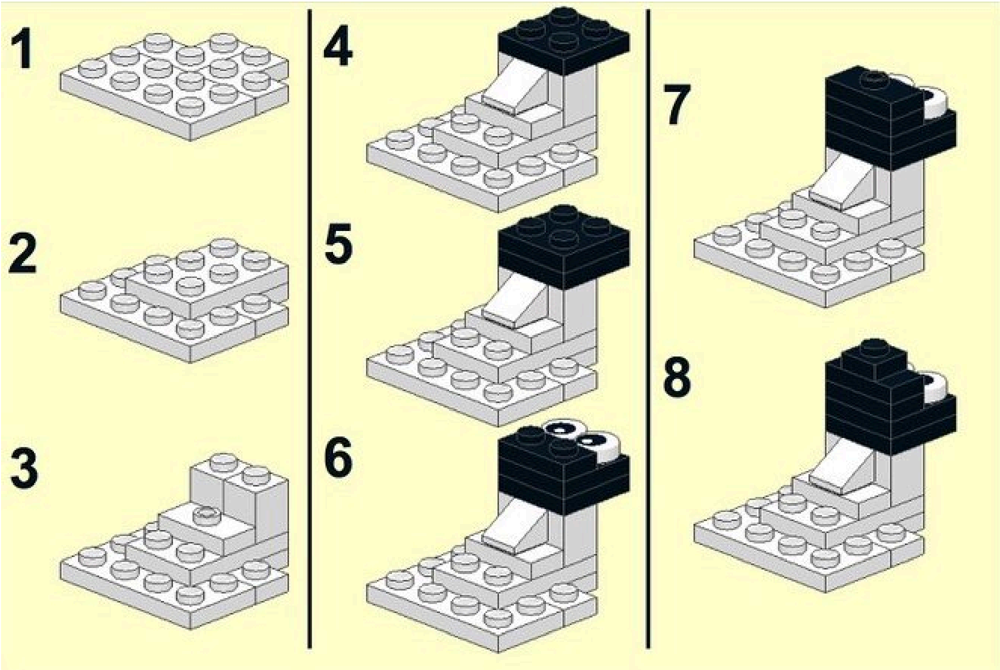
Useful Models

Image: www.flickr.com/photos/brickset/ (CC BY 2.0)

Speaker notes

- But the key is not to worry about those problems at the beginning. as we learn more, we'll see that our models can be more flexible, to represent a wide range of real world systems
- We'll also learn how to fit our models to data, and that's going to make them really useful.

Probability is Rules for Putting Pieces Together



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Speaker notes

- This week, we're starting with the foundation. probability theory
- This is really the nuts and bolts, how do the pieces of statistical models fit together.
- A strong understanding of probability will be useful to you through this course and through your career as a data scientist.

Unit Plan

Speaker notes

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Necessary Foundation

- A precise definition of probability
- How mathematicians build from a set of axioms to useful properties
- How to connect these properties to problems that we want to solve

Speaker notes

No notes on this slide.

Reading Assignment

Reading Assignment

Throughout this course, we intersperse reading, lectures, and work that you complete as a bundled *async* . This way, you can very quickly move from

- An introduction to a concept
- An explanation of its use
- A proof of that concept
- An application using data or problem solving

  We have designed the course so that you can complete the async, reading, and lecture activity components over a few self-contained study sessions.

Speaker notes

No notes on this slide.

Reading Assignment (cont.)

Read the following sections:

- Introduction, page xv - xviii

Speaker notes

- What do we actually say in this?

Probability is Reasoning Under Uncertainty

Speaker notes

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Probablity is A System of Reasoning

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  "Essentially, the theory of probability is
nothing but good common
  sense reduced to mathematics. It provides an
exact appreciation of
  what sound minds feel with a kind of
instinct, frequently without
  being able to account for it.”

  – Pierre-Simon Laplace
```

Speaker notes

- How would you define probability? Is it something that you encounter in your day-to-day life? A measure of randomness in the world?
- Laplace was a famous mathematician, who developed what we call the Bayesian interpretation of probability. Let’s read what he said in 1814.
- Laplace believed that, as humans, we are natural processors of probability. We think in probabilities, we use probability when we speak to each other.
- So of course we think of probability as existing out in the natural world.
- But, I’m here to tell you that probability is actually a model.
- That’s right, the randomness that we perceive around us is a model that we use to make sense of the world.
- Let me try to convince you with a few examples

Probability is A Model of the World

Apocryphal quotations

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``_All models are wrong; some are useful._'' \\hspace{1em} —George Box
```

Probability is:

- A mathematical construct
- A model or abstraction of the world

Speaker notes

- But I'm here to tell you that probability is just a model. It's a very useful model - one that we can make mathematically precise. But it's a model of the world.
- Let me try to convince you with a few examples.
-

Probability is a Model - Example 1

Speaker notes

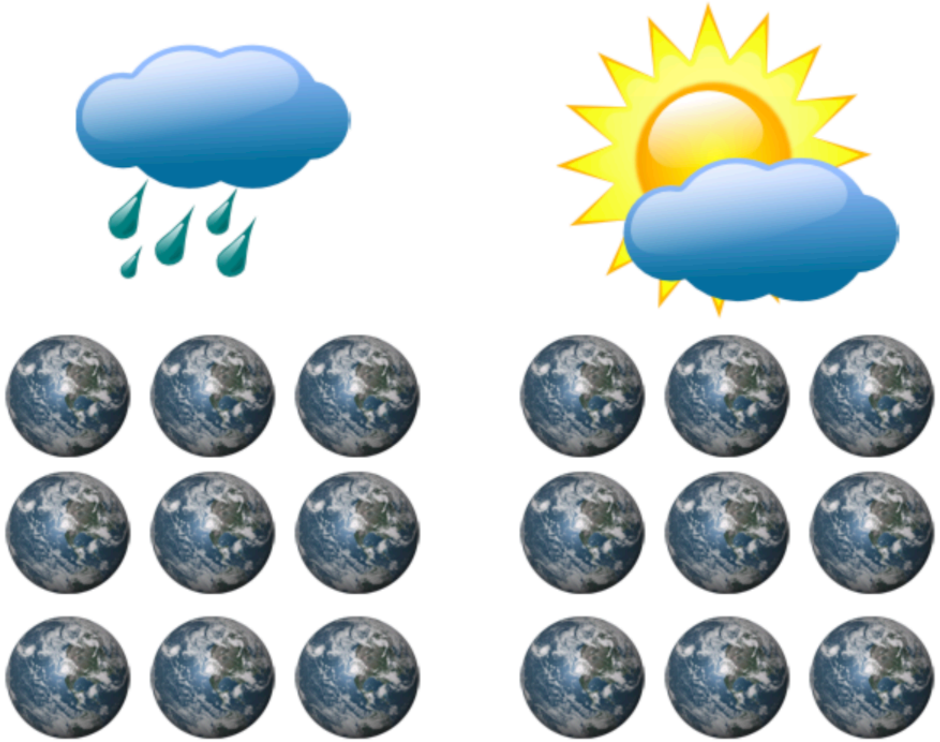
- First, say that you go to the doctor with some abdominal pain.
- The doctor tells you that you have a 80
- But is there any randomness in whether you have kidney stones?
- If you walk out of the office and come back the next day, will you have kidney stones 80
- Of course not. There’s no randomness, you either have kidney stones or you don’t.
- The doctor is using probability in what we call it’s subjective sense – as a strength of belief.

Probability is a Model - Example 2

Speaker notes

- Next, you look at the weather forecast and you see that there’s a 50
- Is there any randomness in whether it will rain? It’s a future event, so you might think maybe.

Probability is a Model - Example 2 (cont.)



Speaker notes

- But what if you made 100 copies of the earth, and arranges the atoms in each one perfectly in exactly the same position with the same speed. would it rain in your location in half of them?
- Probably it would rain in all of them or none. if we could perfectly model how all the atoms in the atmosphere interact, we'd know for sure whether it will rain.
- Again, there's no real randomness - probability is just a model for the world.

Probability is a Model - Example 3

Speaker notes

- Finally, what about the most classic random event of all
  - the coin flip.
- Everyone knows that a coin lands heads 50
- But is that true? What if you could flip the coin in exactly the same way each time.
- And make sure every single atom in the air was moving in exactly the same way.
- You would probably find that it lands on the same side each time.
- Really, the only people who work with true randomness are quantum physicists. The rest of us are using probability as a model.

An Axiomatic Approach

Why take an axiomatic approach?

- Define a minimal set of statements that are consistent with the world we observe
- Deduce statements that must then be logically entailed

Thomas Kuhn's Philosophy of Science

- <1-> Axiomatic statements
- <1-> Intermediate theories
- <1-> Testable hypotheses

Speaker notes

- We already have an intuitive model for probability in our heads, but that's not enough to build statistical models - we need a mathematically precise system.
- How do we create one? We distill our intuition into a small set of axioms. This is what mathematicians do…
- Suppose that you're rolling die – you could observe that the P(1I2) is just P(1) + P(2). You could write that down as a fact.
-
- Instead, we need to identify axioms that explain the probabilities of many different events. induct up to a higher set of principles – the pieces of the model that we need, that are used to repredict to events.
- You might find this frustrating – you might want to study this like it is an observational science; counting lions or cosmic rays. But, it isn't!

Learnosity: Set Theory Diagnostic Quiz

Speaker notes

No notes on this slide.

Set Theory

If you already understand the following notation and can solve the problem, we recommend that you skip this sequence and go straight to the next segment about partitions.

Notation:

A and B are sets.

Simplify the expression: $(A \cap B) \cup (B \setminus A)$

Speaker notes

- To understand the upcoming definitions, you will need to be familiar with basic set theory. This sequence is a quick review of the set operations we will be using most.
- Probably more notation to add, let's keep an eye out and add as we need it.

The Set Operations, Part 1

Speaker notes

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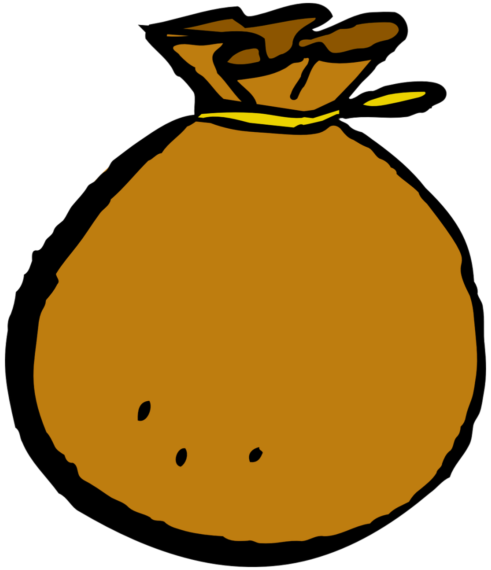
Visualizing a Set



- A set contains objects.
- Items don't have an order.
- Items can't repeat.

Speaker notes

- What is a set? I'm not going to give a precise definition, set theory is interesting but it's pretty far down into the foundations of mathematics.
- What you need to know is that sets contain objects.
- Imagine a bag, so there's no order inside the set.
- An object is either in the set or outside.
- Also, an object can't be in the set twice. No repetition - you're in or you're out.

Elements of Sets

Defining a new set:

- Elements go in curly braces.
- A colon means “such that”.

  Specifying elements:

- ’ $\in$ ’ means ’“is an element of.”
- ’ $\notin$ ’ means “is not an element of.”
- ’ $\emptyset$ ’ means the empty set.

Speaker notes

- How do you write down what elements are in a set?
- To start with, there are two common techniques for building a brand new set
- First, we can just make a list of all the elements inside curly braces.

Subsets and Supersets

Speaker notes

- A subset is set that's contained inside a set
- The larger set is called a superset.
- We take every element that's in at least one of the sets.
- We can write this:
- Most of the time, when we say subset, we mean that A is inside B but A could be exactly the same as B.
- If you ever really need to say that A is a subset and actually smaller, you can add a slash through the bottom line. that's called a strict subset or proper subset.

Set Union

Speaker notes

- A set union means we're merging sets together
- We take every element that's in at least one of the sets.
- We can write this:
- Remember that: Union means OR
- In this case,

Properties of Unions

- Symmetry: $A \cup B = B \cup A$
- Association: $(A \cup B) \cup C = A \cup (B \cup C)$
- $A \cup A = A$
- $A \cup \emptyset = A$

Speaker notes

- I don't want to prove these, becaus
have a lot to cover. they all follow ve
quickly from properties of or, as a lo
operator. I'll prove one to give you a
example
- $x \in (A \cup B) \cup C \Leftrightarrow x \in (A \cup B)$ OR x
- $\Leftrightarrow x \in A$ OR $x \in B$ OR $x \in C$
- $\Leftrightarrow x \in A$ OR $x \in (B \cup C) \Leftrightarrow x \in A \cup$
- Like I said, we're really down in the
foundations.

The Set Operations, Part 2

Speaker notes

No notes on this slide.

Set Intersection

Speaker notes

- A set intersection means we want only the common elements
- We take every element that’s in BOTH of the sets.
- We can write this:
- Remember that: Intersection means AND
- In this case,
- What if there is nothing in the intersection at all?
- If

Properties of Intersections

- Symmetry: $A \cap B = B \cap A$
- Association: $(A \cap B) \cap C = A \cap (B \cap C)$
- $A \cap A = A$
- $A \cap \emptyset = \emptyset$

Speaker notes

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Distributing Union and Intersection

Speaker notes

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Set Difference

Speaker notes

- A set difference means we want the elements in one set but NOT the other.
- It’s like subtracting one set from another.
- We can write this:
- In this case,

Distributing Set Difference

Speaker notes

- There are also ways to distribute set differences over the other operations.
- It can be hard to remember these all at once, but let me highlight this first one, which we’re definitely going to use.

Set Complement

Speaker notes

- We we're doing probability theory, we're going to do all of our work inside a big set, which we call the sample space
- In this case, we have a special name for everything not in a set, A - we call it the complement
-
- The complement is just a set difference,
- In this case,

Set Partitions

Speaker notes

No notes on this slide.

Set Partitions

Definition 1.1.12: ***partition***

```
A set of sets $A_{1}, A_{2}, \dots, A_{n}$ is a _partition_ of set $S$, if $A_{1}, A_{2}, \dots, A_{n}$ are
nonempty and
pairwise disjoint, and if $S = A_{1} \cup A_{2} \cup \cdots \cup A_{n}$.
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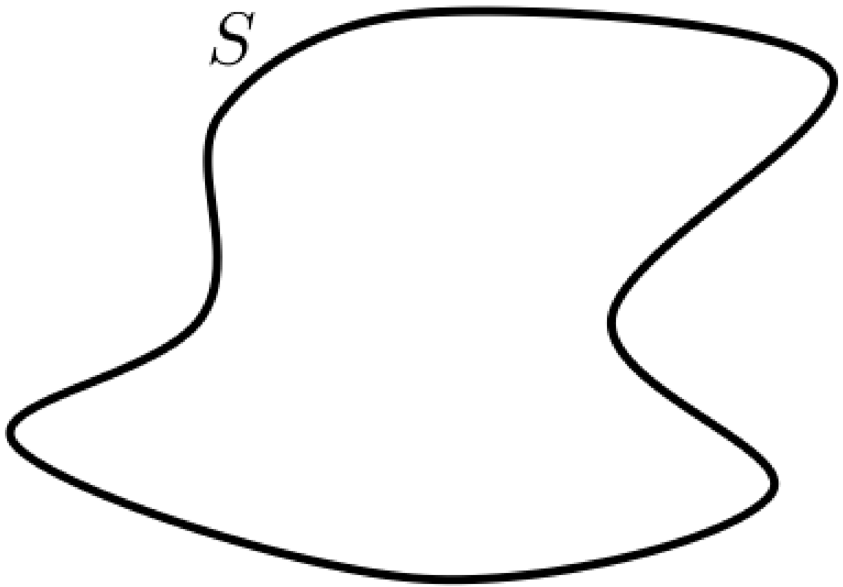
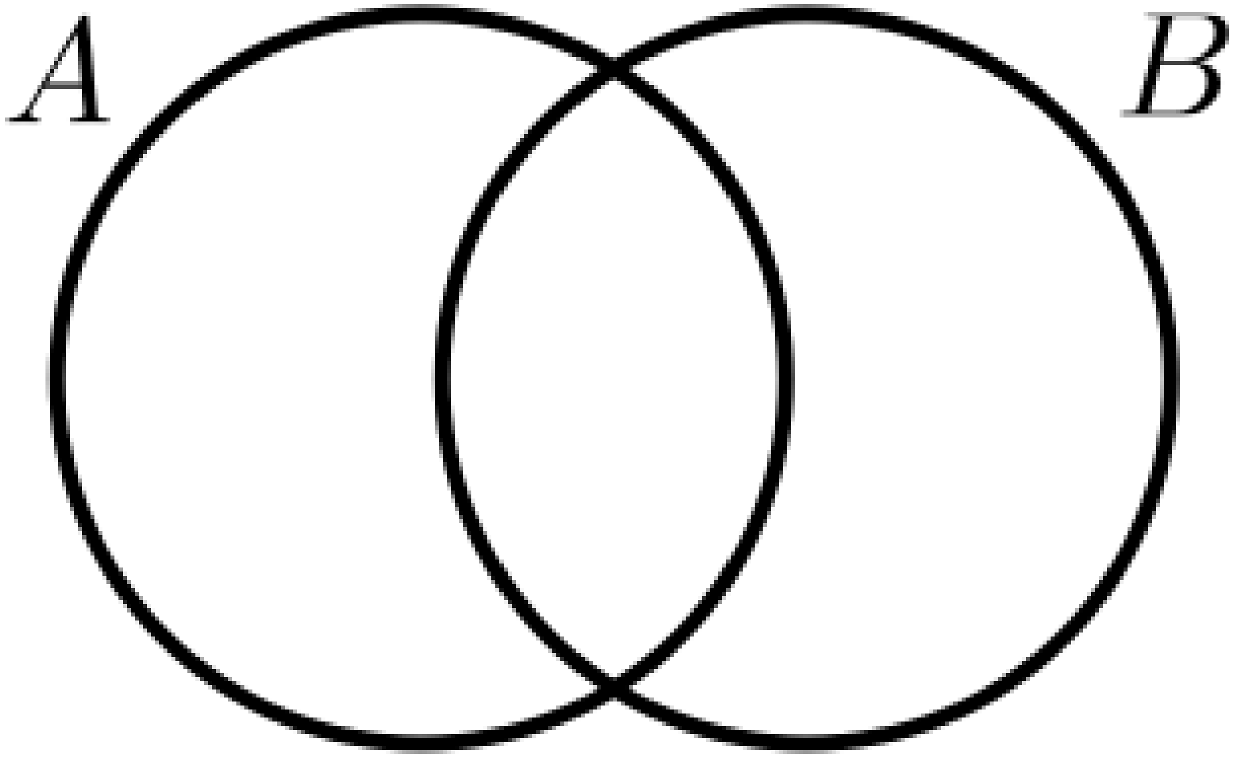


Speaker notes

- While we're talking about sets, there's an idea that's really useful, which is dividing a set up into a partition. Let's read exactly what a partition is…
- Here's a set

Partition Rule 1

Given a set A and another set B , a partition of A is…

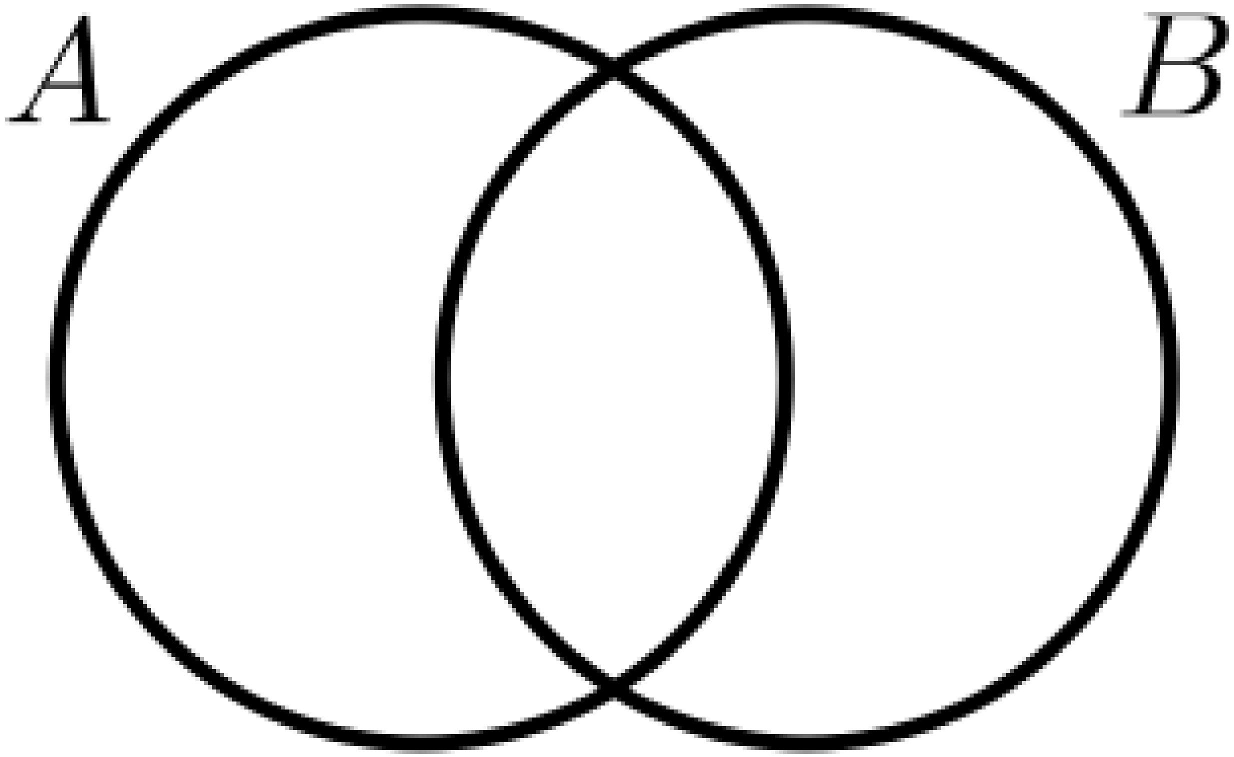


Speaker notes

- Can you see the partition? Here's o the other.
- This one is A minus B, this one is th quickly prove it…
- $A = A \cap (B \cup B^C) = (A \cap B) \cup (A \cap$

Partition Rule 2

Given a set A and another set B , a partition of $A \cup B$ is…



Speaker notes

- Here's a slightly harder problem, wh
  want to partition the entire union?
- This time, we can break it apart into
  pieces. Same as last time, plus also
  minus A.
- Let's quickly prove it…
- Let's just apply the previous relation
  twice!
- $A \cup B = (A \cap B) \cup (A \setminus B) \cup (A \cap B)$
- Guess what, A intersect B is in there
  if you take a union of a set with itsel
  get the same set, so we can just ren
  one of them and we're done.

Call to Reading

- Chapter 1, from page 3 to the top of page 8, stopping before *Theorem 1.1.1*

Speaker notes

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Probability Spaces

Speaker notes

No notes on this slide.

Fundamental Components of Probability Space

> ⓘ A Probability Space
>
> - A **sample space** , denoted as Ω , is the collection of *all* possible outcomes.
> - An **event space** , denoted as S , is made up of sets of outcomes.
> - A **probability measure** is a function that maps events to numbers in $[0, 1]$. % include squiggly picture

Speaker notes

- To build statistical models, we need a mathematical description of probability.
- We need to take our intuition for what a system with uncertainty means, and encode it into a mathematical object. that object will be what we call a probability space.

Sample Spaces

Choosing sample space Ω to represent…

- A die % which is discrete
- A dartboard % which is continuous
- A survey of 5 U.S. senators
- A spoken conversation

Speaker notes

- Let's start with the sample space. where do we get the sample space? well we have to design it to represent the real world. Let's give some examples.
- A die –
- A dartboard –
- A survey of 5 U.S. senators –
- Suppose you're doing language translation –
- Question: can you apply a probability measure directly to the set of outcomes? unfortunately, no. for simple ones you can, but what about the dartboard. there are an infinite number of locations, and the probability of each one is just zero. but a function that's zero everywhere doesn't help us model this.

Events

An event is a set of outcomes.

- Rolling an even number. % Rolling a 4.
- Hitting a bullseye. % Getting five dirty hippies in your survey: ${ {}, $ %
  $\{\text{Mike Rivera, Dawn, Harmony, Karma, Leaf}\}, \ldots \}$
- Starting a sentence with ‘Um’.

Speaker notes

No notes on this slide.

Forming New Events

$A = \{1, 2\}, B = \{2, 4, 6\}$ - A OR B: - A AND B:
- NOT A:

Speaker notes

No notes on this slide.

Event Spaces

Intuition for an event space

- S represents all possible events - However, there are technical reasons that some sets of outcomes must be excluded (non-measurable).

Requirements for a σ-algebra - Nonempty: $S \neq \emptyset$ - Closed under complements: If $A \in S$ then $A^C \in S$ - Closed under countable unions: if $A_1, A_2, ... \in S$ then $A_1 \cup A_2 \cup ... \in S$

Speaker notes

- So Mathematicians said: we can't put all possible sets in the event space, let's just make sure that any set we need for a regular analysis is in there. So they came up with a list of guarantees and the technical term for these is a
- You can think of this as making sure that any way you can combine events together, the answer will always be an event, so it will have a probability. You can't accidentally fall out of the event space.

Probability Measure

Definition of a probability measure

A function $P: S \rightarrow \mathbb{R}$ is a
probability measure if:
- __Axiom 1.__ There are no negative probabilities: $\forall A \in S: P(A) \geq 0$

- **Axiom 2.** The probability that *some* outcome occurs is one: $P(\Omega) = 1$
- **Axiom 3.** Probabilities "add up": If $A_1, A_2, ...$ is a countable set of disjoint events, then $P(A_1 \cup A_2 \cup ...) = \sum_i P(A_i)$

Speaker notes

- Finally, the last element is the probability measure itself. To be a valid probability measure, a function has to satisfy three properties.
- Where do these come from? We're building a model here. we're taking properties from our intuition and trying to write them down using math.
- First, there are no negative probabilities. that just makes sense to us, right? what would a negative probability mean?
- Next, the probability of the entire outcome space is 1. That should make sense too, because some outcome must happen. the state of the world will be something.
- Finally, the addition rule. Probabilities add up when we combine events, as long as they are disjoint. The probability of rolling a 1 or a 2 is the probability of rolling a 1 plus the probability of rolling a 2. But in standard probability theory that

Probability Measure

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Speaker notes

- Looking back at this list of axioms, you might wonder, why only 3?
- What happens if we take a union, but events aren't disjoint? Don't we need a property about that?
- That is an important property, but we don't need it here, because we can derive it from just these three. That's what we'll do next.
-

Learnosity Concept Check

Speaker notes

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You run an experiment in which you flip 3 coins, and each one lands heads or tails. Your sample space can be written: $\{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$.
Assume all outcomes have equal probability.

- (Assuming that you include all possible sets) what is the size of the event space?
- How many outcomes are in the event that the first coin lands heads?
- If A is the event that the first coin lands heads, and B is the event that the second and third coin land the same way, what is the probability of $A \cup B$

Speaker notes

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A Single Coin Example

Speaker notes

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-

Learnosity: A Double Coin Example

Speaker notes

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Reading: Properties of Probability

Speaker notes

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Reading: Properties of Probability

Read page 8 through the end of page 10.

Speaker notes

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Deriving Properties of Probability

Speaker notes

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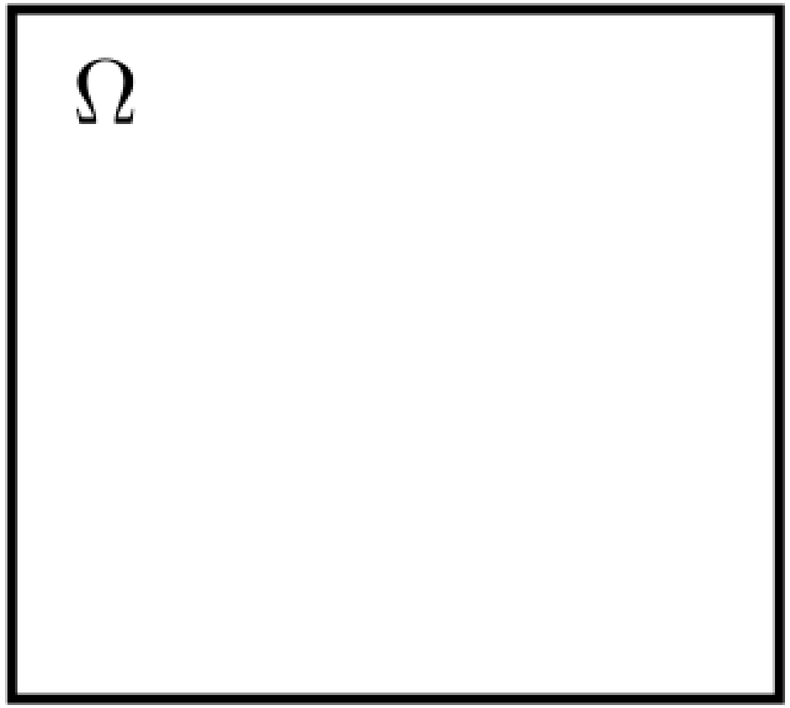
A Helpful Analogy

Probability $\iff$ Area



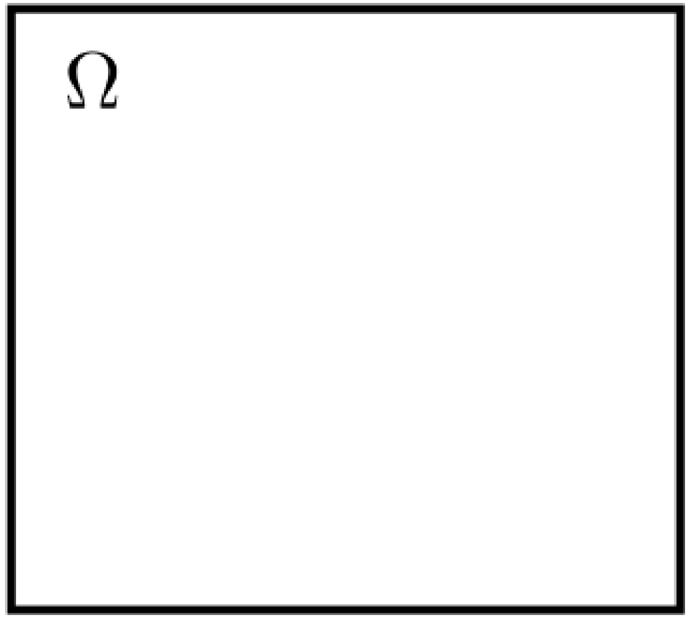
Speaker notes

- The amazing thing about probability is that with just 3 axioms, you can prove all the other properties that you expect based on your intuition. But it can be tough to find a solution just by staring at equations.
- Here's an analogy that may help you.
- Probabilities are a lot like areas. Actually, area is also a type of measure in mathematics - most of the structure is the same.
- If you just have a few events, it can help to draw them as areas. That's not a proof, but it can give you an idea of how they behave.
- Let's think about the sample space as a square dartboard. We'll define its area to be 1. I'm really bad at darts so every point is just as likely as any other point.
- An event is some shape on the dartboard. so the probability that the dart lands in the shape is its

The Subset Subtraction Rule

Given $A, B \in S$, $A \subseteq B$. Compute $P(B \setminus A)$.

Ex: A = being over 7’, B = being over 6’.



Speaker notes

- Let’s start with a simple property, if
a subset of event B, what’s the prob
B.
- For example, if A is the event of bei
the event of being over 6 foot tall. h
probability of being between 6 and
- I’ll draw a circle for B and a circle fo
axioms, we need disjoint sets. A an
know how to partition
- $B = (B \setminus A) \cup (A \cap B) = (B \setminus A) \cup A$
- Also, by axiom 1,

The Complement Rule

Given $A \in S$. Compute $P(A^C)$.

Speaker notes

- Given the subtraction rule, you can very quickly solve for the complement rule
- Note
- subtraction rule
- Axiom 2:
- $P(A^C) = 1 - P(A)$

Speaker notes

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Learnosity Concept Check - Addition Rule

Speaker notes

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Here is a proof of the addition rule, select the best justification for each line.

$$
\begin{aligned}
A \cup B &= (A \setminus B) \cup (A \cap B) \cup (B \setminus A) \\
P(A \cup B) &= P(A \setminus B) + P(A \cap B) + P(B \setminus A) \\
&= P(A \setminus B) + P(A \cap B) + P(B \setminus A) + P(A \cap B) - P(A \cap \\
&= P\Big((A \setminus B) \cup (A \cap B)\Big) + P\Big((B \setminus A) \cup (A \cap B)\Big) - P(A \\
&= P(A) + P(B) - P(A \cap B)
\end{aligned}
$$

Options (to be scrambled):

- Partition Rule 2
- Axiom 3, countable additivity.
- Because adding something and also subtracting it doesn't change anything.

Speaker notes

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Learnosity Concept Check - Applying the Addition Rule

Speaker notes

No notes on this slide.

Applying the Addition Rule

A station along Route 66 sells gas and postcards. The probability that a customer buys postcards is .4. The probability that a customer leaves without buying anything is .3. The probability that the customer buys both gas and postcards is .2. What is the probability that the customer buys gas?

Answer: .5

Speaker notes

No notes on this slide.

Conditional Probability

Conditional Probability, Part I

Conditional probability is a rescoping of the sample space from Ω —every possible outcome—to a smaller set called the *conditioning set* - What is the probability that a student on campus is a volleyball player (V)? - What is the probability that a student on campus is a volleyball player (V), given that they are 6’3’ (T)’?

Uses information to produce a *better* statement of the likelihood of an event

Speaker notes

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-

Think of a Dartboard…

Speaker notes

- Let’s build up some intuition by thinking about the dartboard analogy.
- Event V for volleyball player is an area on the board.
- If all you know is that the dart hit the board randomly, the probability that it’s in V is the area of V divided by the area of the entire sample space (which we assume to be 1).
- Now what if I tell you that the dart landed somewhere in T, meaning tall? What does your intuition say?
- Well, now you don’t care about the rest of

Conditional Probability, Part II

Conditional probability

```
For events $A, B \in \Omega$ with $P(B) > 0$, define the _conditional
  probability_ of $A$ given $B$ to be
\[
  P(A|B) = \frac{P(A \cap B)}{P(B)}
  \]
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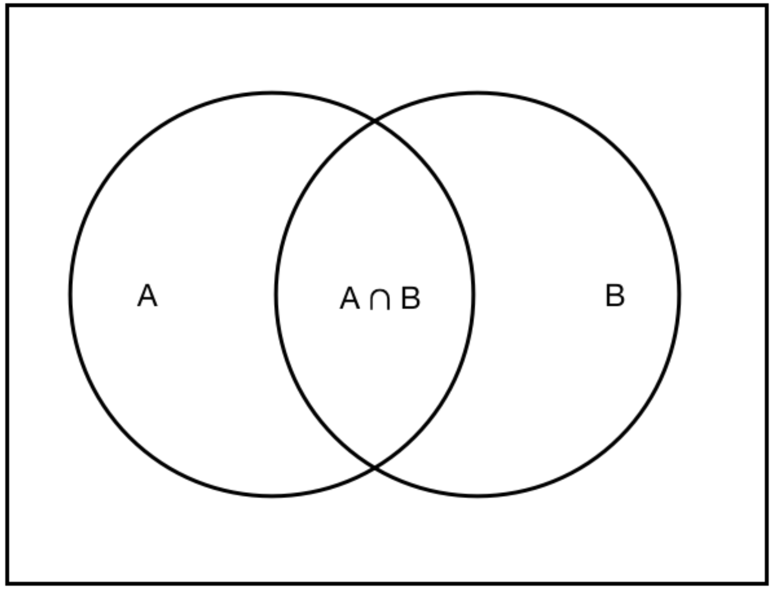


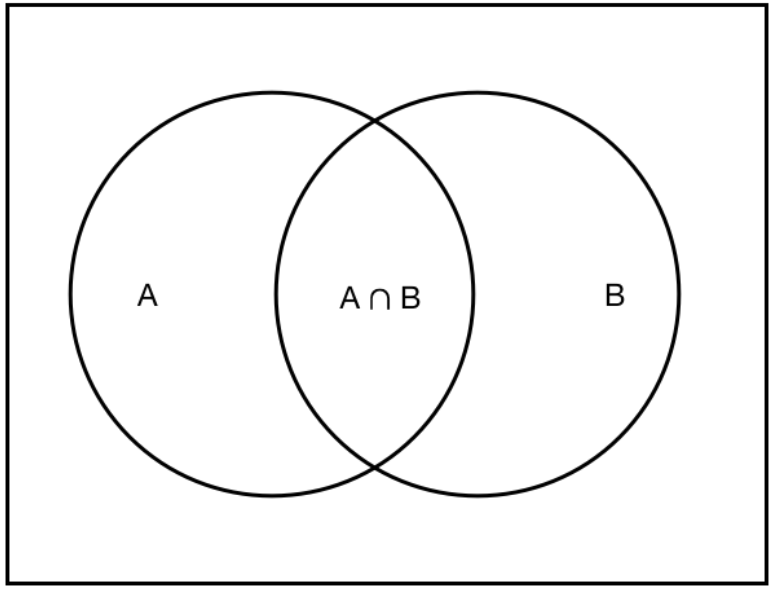
Speaker notes

- Here’s the formal definition. the conditional probability of A given B is a ratio.
- On the top we have the intersection. on the bottom we have all of B

Conditional Probability, Part III

Conditional probability

```
For events $A, B \in \Omega$ with $P(B) > 0$, define the _conditional
  probability_ of $A$ given $B$ to be
\[
  P(A|B) = \frac{P(A \cap B)}{P(B)}
  \]
```



Speaker notes

- The idea is we're dropping everything in the sample space outside of B - it's not relevant, and then we have to rescale the probability so everything adds up to 1.

Conditional Probability, Part IV

Multiplicative Law of Probability

```
Rearranging the conditional probability statement minimally
produces the _Multiplicative Law of Probability_.

For $A, B \in \Omega$ with $P(B) > 0$,

\[
  P(A|B)P(B) = P(A \cap B) \pause = P(B|A)P(A)
\]
```


Speaker notes

-
- This is an important relationship. It gives us an intuitive way to compute the probability that A AND B occur.
- First, you write down the probability that A occurs. then, you assume that A has occurred, and you multiply by the probability, which is conditional, that B also occurs.
-

Learnosity Check

Speaker notes

No notes on this slide.

Suppose that the probability that a student plays volleyball is .4, the probability that the student is tall is .5, and the probability that the student is both tall and plays volleyball is .3.

What is the conditional probability that the student plays volleyball, given that they are tall?

Answer: .6

Speaker notes

No notes on this slide.

Reading: Bayes' Rule

Speaker notes

No notes on this slide.

Reading: Bayes' Rule

Read from the top of page 11 to the end of page 13.

Speaker notes

No notes on this slide.

Total Probability

Speaker notes

No notes on this slide.

Total Probability, Part I

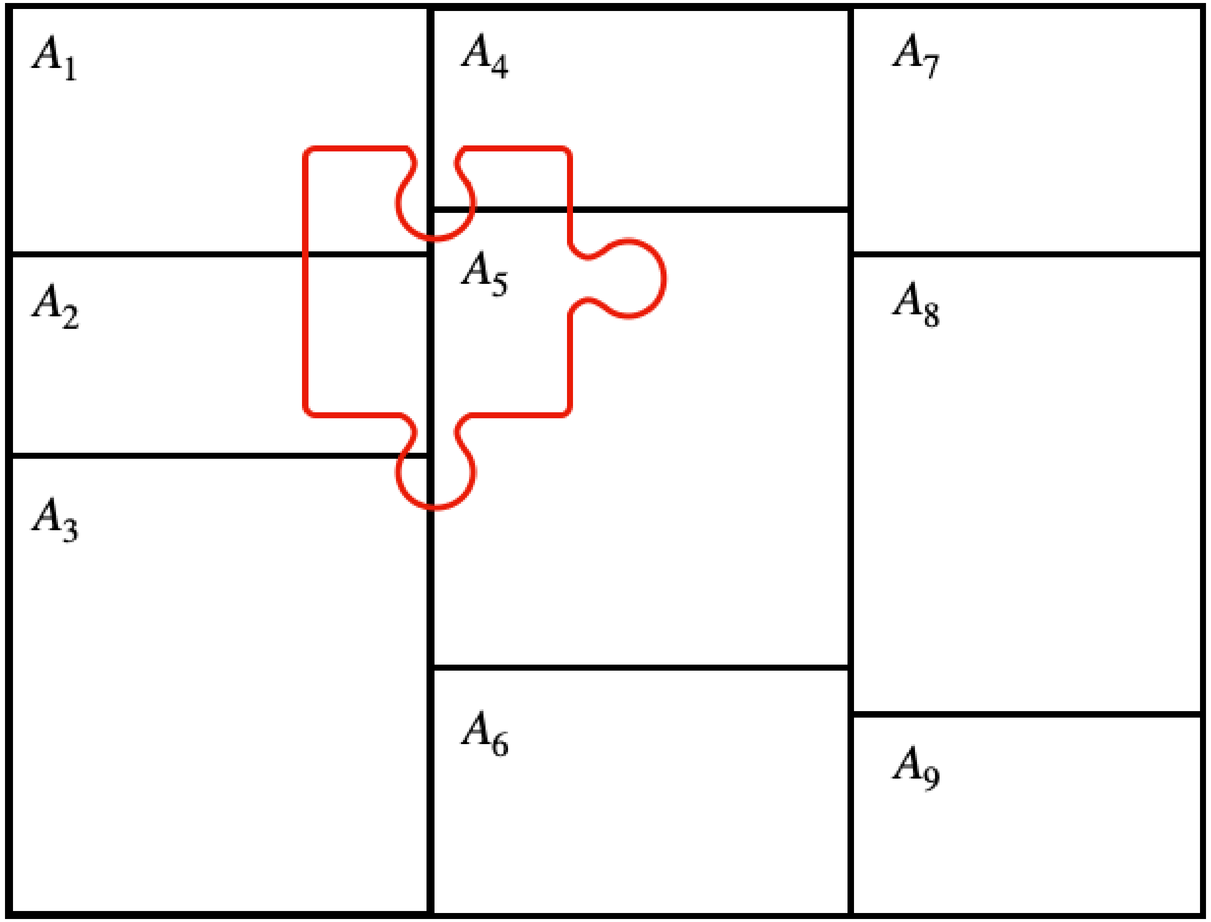
The Law of Total Probability

```
If$\ A_{1}, A_{2}, \dots, A_{n}$ is a partition of $\Omega$, and
if $B \in \Omega$, then
\[
  P(B) = \sum_{i} P(B \cap A_{i})
\]
If there is a positive probability for each partition $A_{i}$,
then by the _Multiplicative Law of Probability_\[
  P(B) = \sum_{i} P(B | A_{i})P(A_{i})
\]
```


Speaker notes

- In a lot of probability problems, you need to compute the probability of some complicated event.
- That can be hard to do directly, and you want some way to break down the problem.
- There is a famous technique that helps in a lot of cases. it’s called the law of total probability.
- The idea is we can use a partition of
- Let’s see a picture and then read the law carefully…

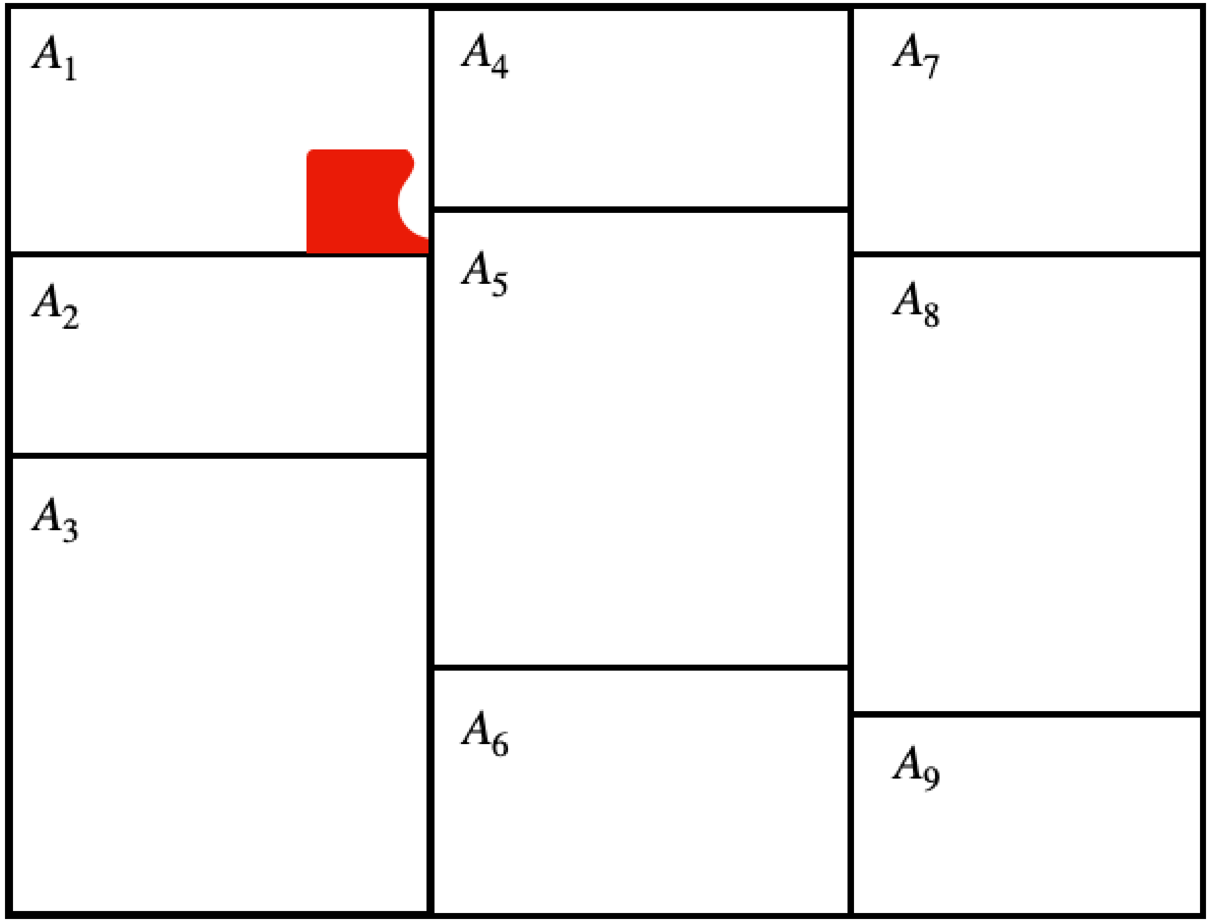
Total Probability, Part II



Speaker notes

No notes on this slide.

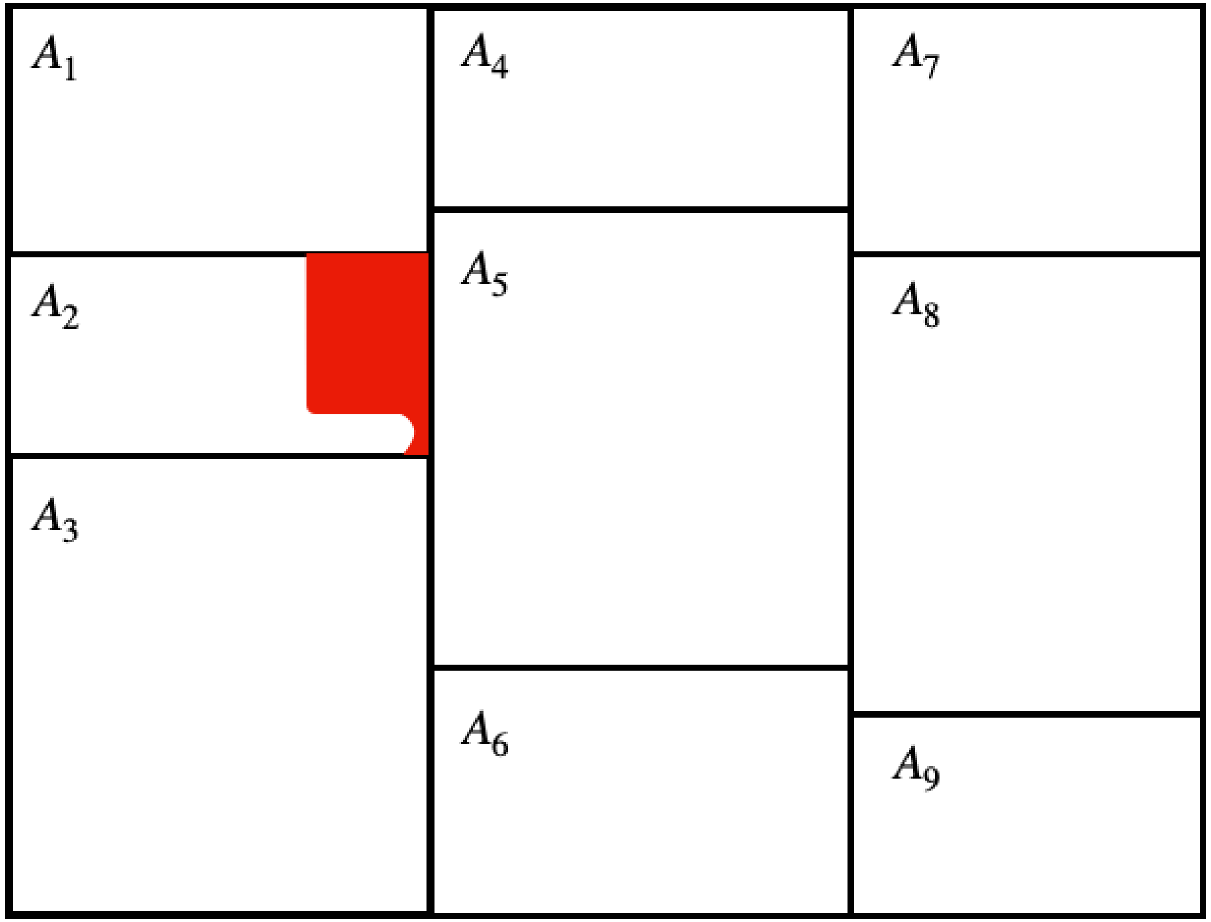
Total Probability, Part III



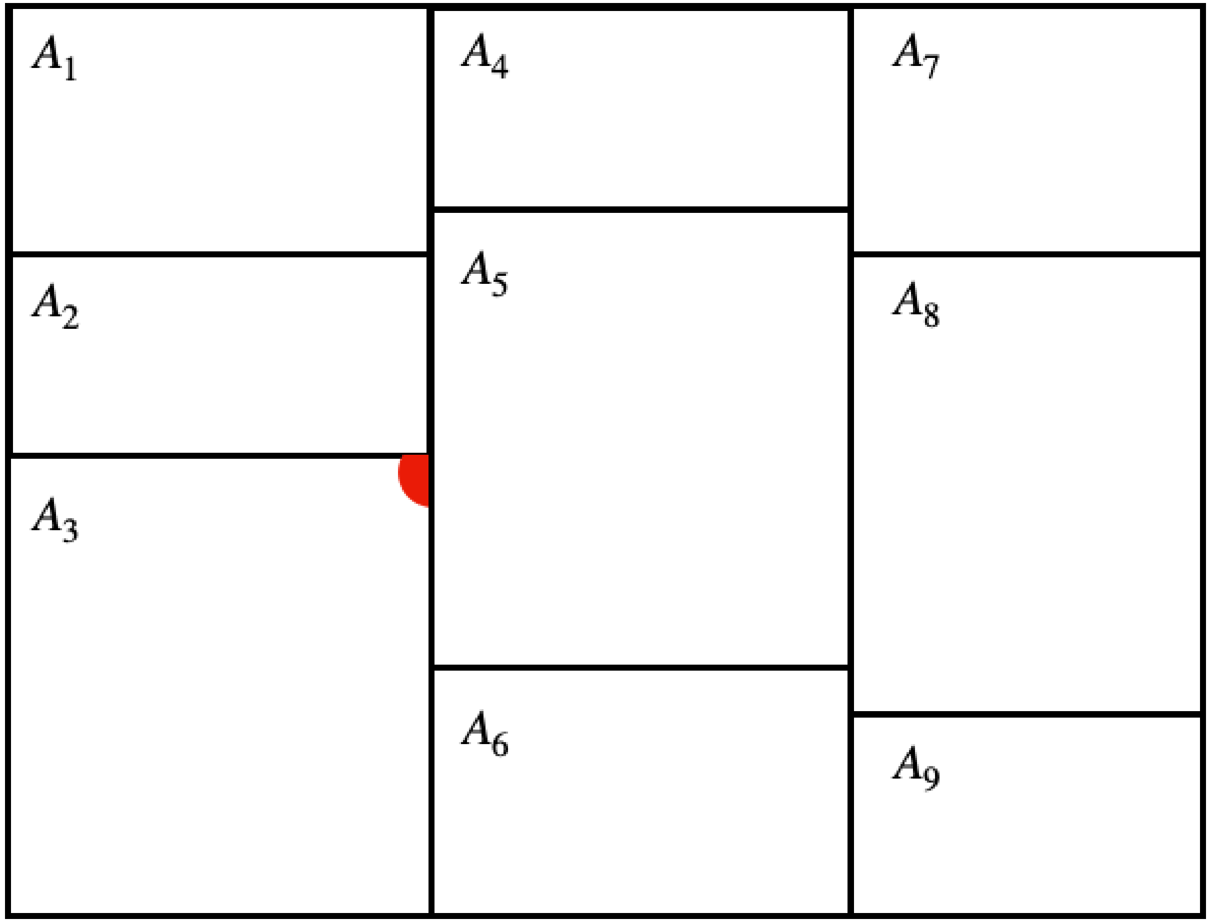
Speaker notes

No notes on this slide.

Total Probability, Part IV



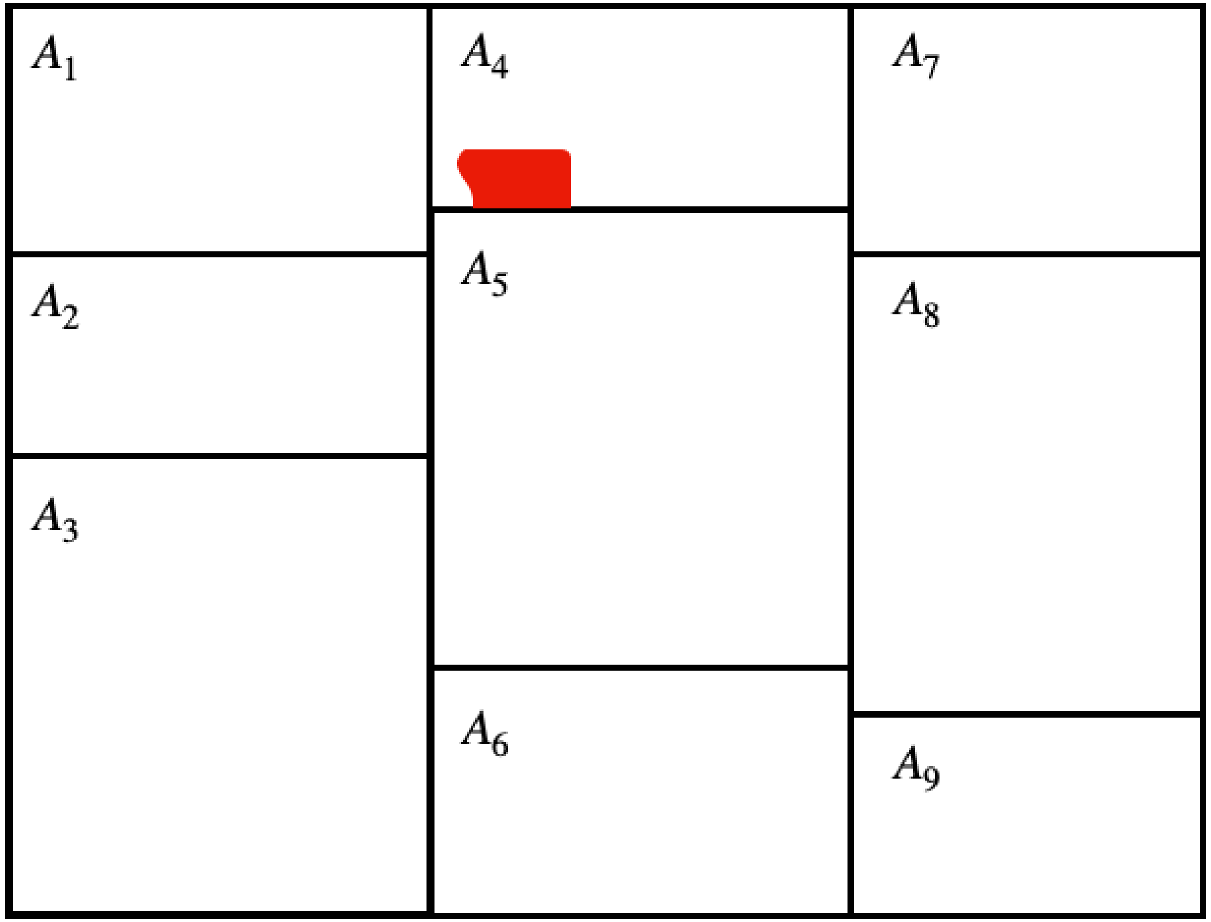
Total Probability, Part V



Speaker notes

No notes on this slide.

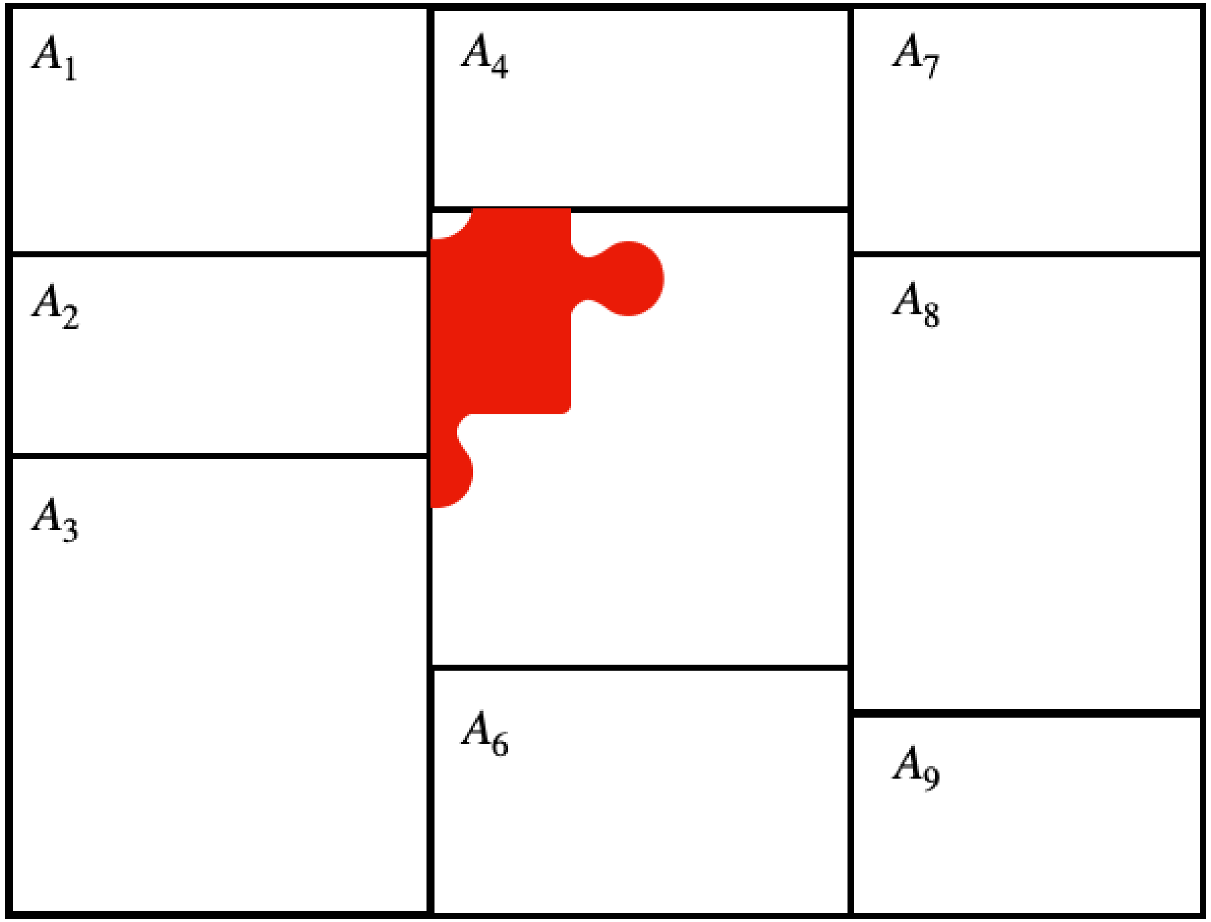
Total Probability, Part VI



Speaker notes

No notes on this slide.

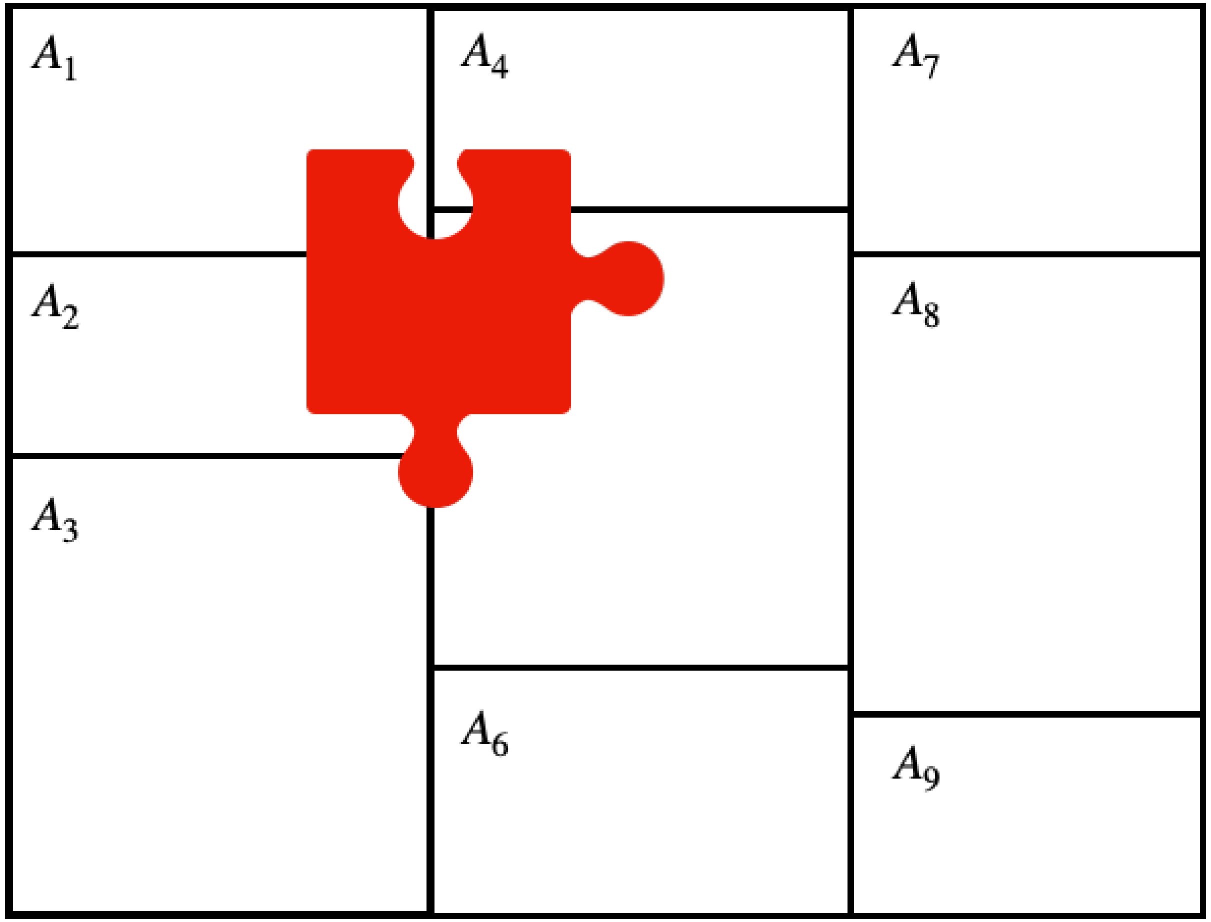
Total Probability, Part VII



Speaker notes

No notes on this slide.

Total Probability, Part VIII



Speaker notes

No notes on this slide.

Total Probability, Part I

The Law of Total Probability

```
If$\ A_{1}, A_{2}, \dots, A_{n}$ is a partition of $\Omega$, and
if $B \in \Omega$, then
\[
  P(B) = \sum_{i} P(B \cap A_{i})
\]
If there is a positive probability for each partition $A_{i}$,
then by the _Multiplicative Law of Probability_\[
  P(B) = \sum_{i} P(B | A_{i})P(A_{i})
\]
```


Speaker notes

- Now let’s carefully read the law of total probability.
- The probability of B is a sum, if you look inside the sum, what we do is we intersect B with some
- To finish the law, we can then apply the multiplication rule. we write each of these joint probabilities as a product. First, we take the probability we’re inside
- That’s the law of total probability. It can be quite useful, if you can think of a good partition.

Bayes' Rule

Speaker notes

No notes on this slide.

Conditional Inverses

Does $P(X|Y) = P(Y|X)$?

Example:

- X = Visited Berkeley
- Y = Attend MIDS

Speaker notes

- We know about conditional probability. Let's say we have two events X and Y. Are these two conditional probabilities the same?
- In fact, the answer is usually no. here's an example to convince you.
- There is an easy way to get from one of these probabilities to the other. it's the famous theorem known as Bayes' Rule.

Deriving Bayes' Rule

$P(X \cap Y)$

Speaker notes

- Deriving Bayes' Rule is easy. We start with the probability of X and Y. Then wewrite the multiplication rule two ways.
- $P(Y)P(X|Y) =$
- Divide:
- That's Bayes's Rule! It's easy to derive if you ever forget it.

Bayes' Rule Example

- A rare disease affects 2 out of 10,000 people.
- The test is right 99 % of the time for healthy people.
- The test is right 100 % of the time for sick people.

Given that you get a positive test, what is the probability that you have the disease?

- Let D be the event you have the disease
- Let T be the event the test is positive.

Speaker notes

- Why is Baye's rule important? It can tell us how to adjust probabilities of hypotheses in response to evidence.
- Let's work through a problem together. This is a very classic example for Bayes' Rule.
- You can see that this sounds quite scary, right? the test is correct 99
- Write down what we know:

Bayes' Rule Example Cont.

We know:

$$P(D) = .0002, P(D^C) = .9998$$
$$P(T|D) = 1, P(T|D^C) = .01$$

We want: $P(D|T)$

Speaker notes

- $P(D|T) = \frac{P(T|D)P(D)}{P(T)}$
- Still missing
- $P(T) = P(D)P(T|D) + P(D^C)P(T|D$
- $P(D|T) = \frac{P(T|D)P(D)}{P(T)} = \frac{P(T|D}{P(D)P(T|D)+P(D^C}$
- Even though the test is 99

Bayesian Statistics

Let H be the event a hypothesis is true.

- We begin with prior (subjective) belief $P(H)$

  Let D be the data we collect
- Use Bayes' Rule to compute posterior belief, $P(H|D)$.

Speaker notes

- Bayes' Rule is the foundation of an entire school of stats.
- Let me try to compress an entire school into 60 seconds.
- In bayesian stats, we may be interested in some hypothesis. Let H be the event that it's true.
- To a mainstream statistician, that doesn't make any sense, a hypothesis is true or false.
- To a bayesian, probability is used to represent your personal subjective belief. before you see the data, you give the hypothesis a prior probability, P(H)
- Then you see data, D. you can use bayes rule to update your probability to
- There's your 60 second crash course on Bayesian stats.

Reading: Independence

Speaker notes

No notes on this slide.

Reading: Independence

Read section 1.1.4, Independence of Events

Speaker notes

No notes on this slide.

Independence

Speaker notes

No notes on this slide.

Independence

Independence of events

```
Events $A, B \in S$ are _independent_ if $P(A \cap B) =
P(A)P(B)$.
```

Theorem: conditional probability and independence

```
For $A, B \in S$ and $P(B)>0$, $A$ and $B$ are
independent if and only if $P(A|B) = P(A)$.
```


Speaker notes

- Let's take a look at the formal definition of independence. Notice that we use the elegant product equation. This version is a bit more universal. If event have probability zero, there may not be a good way to define conditional probability.
- However, as long as the events have nonzero probability, you can also write independence in terms of conditional probabilities.

Conditional Probability and Information

- Let C be event that our subscriber cancels.
- Let W be the event our subscriber waited 2 hours on the phone. $P(C) < P(C|W)W$ gives us_information_ about C.

%%<— here

Speaker notes

- We introduced conditional probability, and we can think about it in terms of information.
- This is what we usually expect, at least a little bit. one event will give us some information about another event. But it’s not always the case.

Understanding Independence

- Let C be event that our subscriber cancels.
- Let H be the event our subscriber hates cilantro. Is$P(C)$ different from $P(C|H)$?

%%<— here

Speaker notes

- Same situation, but now let H be the event the subscriber hates cilantro.
- Now, does knowing that the subscriber hates cilantro give us any information about whether they cancel?
- That’s pretty farfetched, right? Unless you’re actually sending cilantro to your subscribers - let’s assume that’s not the case - these events have nothing to do with each other. So
- Let’s play with the math.
- So
- Also,

Importance of Independence

Later in the course: In a *random sample* every unit is independent of every other unit.

Survey Example:- Let D_1 be event that the first respondent has a dog. - Let D_2 be event that the second respondent has a dog. -

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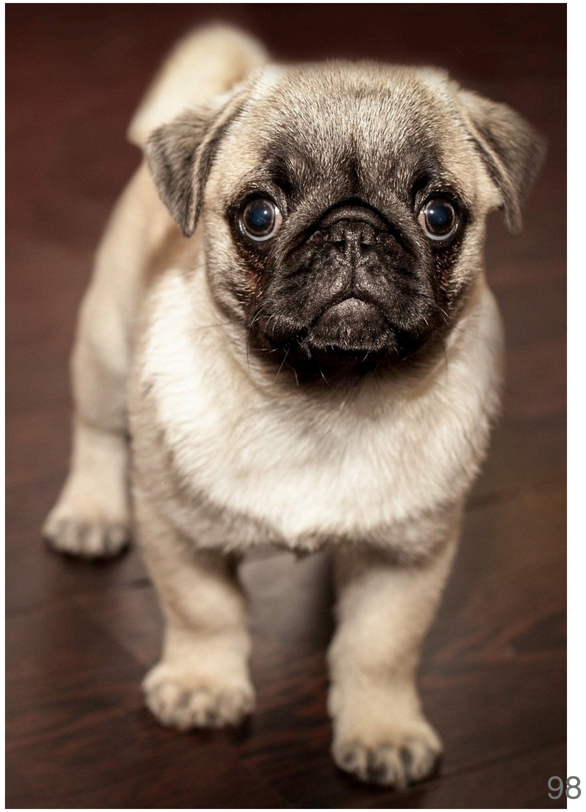

Speaker notes

- My cilantro example is pretty silly. So why do we need independence?
- It's going to become very important when we talk about sampling.
- Think of a survey.
- if
- But what it they are family members? Then the responses of the first one probably does give you information about the second. This makes it much harder to analyze the data. Throughout this course, we always want to be skeptical about whether independence holds, but many of our techniques really will require it to work.

Framing the Following Segment

Speaker notes

No notes on this slide.

Fatal Officer Involved Shootings

We’re going to take on an important, but challenging topic here – the use of police force.

- In particular, we’re going to read a paper that investigates whether office race plays any role in disparate use of force against people of color
- This is an important, durable question and one that at the time that we’re putting the course together is the predominant political issue.

Speaker notes

No notes on this slide.

Fatal Officer Involved Shootings

As faculty, we stand is solidarity with people of color, without exception and without qualification.

- We’re going to read an academic, data-based conversation where the authors come to the conclusion that there is no evidence of differential treatment of people of color by white officers.
- And, we’re going to critically engage with this article, to assess whether the data supports this claim

Speaker notes

No notes on this slide.

Probability and Police Shootings

Speaker notes

No notes on this slide.

standout

Are Black citizens subjected to unfairly violent treatment by law enforcement?

Speaker notes

-
- Let’s start our discussion of probability with this question…
- This is a matter that we care deeply about (as I’m sure many of our students do), but I want to make clear that we want to approach it today from an academic perspective - that means being dispassionate and using data to construct measurements that aren’t biased by our personal convictions.

Johnson et al. (2019)

> **Johnson et al. (2019)**
>
> ```
> Are minority groups subjected to unfairly violent treatment
> by law enforcement?
> ```
>
> 1. Are Black civilians overrepresented in FOIS?
> 2. If so, is this because of racial discrimination by White officers?

- Develop Database that is a near-total recording of all *fatal officer involved shootings* (FOIS) in the US in 2015.
- ` ` `Model'' whether race of civilian is a factor that predicts being involved in a FOIS.

Speaker notes

-
-

Johnson et al. (2019) Conclusions

Study conclusions

- *``We find no overall evidence of anti-Black or anti-Hispanic disparities in fatal shootings.''*
- *``White officers are not more likely to shoot minority civilians than non-White officers.''*

Speaker notes

No notes on this slide.

Probability Translation

Crystal clear probability statement about officer race:

$$P(\text{shot}|\text{minority civilian, white officer}, X) \leq P(\text{shot}|\text{minority civilian, minority officer}, X)$$

Question: Is it possible to justify this statement using only data of officer shootings?

Speaker notes

- The answer to this probability statement informs policy: If true, then take proactive measures in hiring
-
-

Measuring Fairness

- Twenty-six percent of civilians killed by police shootings in 2015 were Black.
- The U.S. population is 12 % Black.
- White and Black civilians have different rates of exposure to situations that might result in FOIS.

Speaker notes

- **We need these numbers**
- We could control for interactions with police. But, there are all kinds of reasons that these interactions – set by policy – are different between groups. The Oakland police are under federal receivership because of their treatment.